

NovusÆxenti (**novus-aexenti**)

Subsystem Identity: Cognitive MoE Reasoning Engine & Dynamic Memory Flywheel

Authority & Reference Specs:

- [A OPERATOR MAP.md](#)
 - [A+ 03 WORKSPACE FILE TREES AND TOPOLOGY SYNTHESIS. this explains a lotmd](#)
 - [PROPOSAL-V2-CORRECTED-WIKI-TERMINOLOGY-AND-s-tier-ALTERNATIVE.md](#)
 - [02 DUMBASS UNIVERSAL MEMORY SPEC.md](#)
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1. Subsystem Scope & Architectural Responsibilities

novus-aexenti houses the on-device cognitive routing and reasoning flywheel for Project NovÆxorpheus:

- **0.8B Triage Router:** Fast-path token classification, prompt categorization, and low-latency dispatch.
 - **9B Executor Model:** Local deep-reasoning agent executing complex workflows on Snapdragon Hexagon NPU / edge runtimes.
 - **Dynamic Memory Tap Integration:** Communicates via OmniRoute (localhost:20128/v1) for continuous memory extraction without inference latency.
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2. Directory Layout

Domain-Specific Subfolders

- **dual_agent_router/** — Fast triage routing configurations, token classification matrices, and dual-model handoff logic.
- **mem0_episodic/** — In-session short-term episodic state, user habit registries, and OB1 static protocol bridges.
- **reasoning_bank/** — Execution traces, active execution paths (**active_execution_paths.json**), failure logs, and crash recovery ledgers.
- **nope_databank/** — Negative-prompt datasets, anti-patterns, rejected trajectories, and failure boundary constraints.

Universal Baseline Subfolders (The 10 Subfolders)

- `raw/` — Ingestion landing zone for raw cognitive traces and unconverted inputs.
 - `clean_md/` — High-density condensed markdown maintaining 100% build context.
 - `wiki_md/` — Compounding living wiki knowledge nodes with bidirectional `[[wikilinks]]`.
 - `skills/` — Procedural execution policies and reasoning skills applied locally.
 - `tools/` — Local executable utilities and CLI helpers.
 - `scripts/` — Subsystem-specific executable shell and Python scripts.
 - `hooks/` — Subsystem lifecycle, git, and sync hooks.
 - `pending/` — Staged tasks, unverified reasoning traces, and backlog items.
 - `audit/` — RLVR verification reports, divergence traces, and test-case outputs.
 - `archive/` — Deprecated models, historical traces, and superseded configurations.
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3. Operational Rules

1. **Locality of Reference:** Skills and tools developed for cognitive triage and reasoning live inside `skills/` and `tools/` in this repository.
2. **First-Move Directive:** Read `MAP.md` to orient -> query `manifest.jsonl` to filter -> load only targeted files into context.
3. **Beginner-Proof Invariant:** All scripts, routing rules, and configurations must be clear, fully functional, and self-documenting.